

Microwave Thermoradiotherapy for Uveal Melanoma

Results of a 10-year Study

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Objective: The purpose of the study is to evaluate clinically the use of microwave-heating (hyperthermia) as an adjuvant to ophthalmic plaque irradiation for treatment of patients with uveal melanoma. Hyperthermia was also used as a radiation sensitizer, allowing for significant dose reductions during ophthalmic plaque radiation therapy.

Participants: In this case series, 48 patients were treated with microwave plaque thermotherapy for uveal melanoma.

Intervention: Microwave treatment, which involved affixing a miniature microwave dish antenna on the sclera beneath the tumor after completion of plaque brachytherapy, was performed. During hyperthermia treatment, the tumor's apex was targeted to receive a minimum of 42 °C for a 45-minute duration. A subset of 38 (79%) were given reduced apical doses of ophthalmic plaque radiation (radioactive isotope of iodine [¹²⁵I] or palladium-103 [¹⁰³Pd]) to an average of 52.6 Gy.

Main Outcome Parameters: Patients were evaluated for visual function, microwave toxicity, radiation oculopathy, eye retention, local tumor control, and metastatic disease.

Results: Patients have been observed for up to 10 years and for an average of 60 months (5 years). To date, there have been 3 cases of postoperative tumor enlargement (growth) for a 93.8% local control rate. Two patients were lost to follow-up. Seven eyes have been enucleated: three due to neovascular glaucoma, one due to uveitic neovascular glaucoma, and three due to progressive tumor enlargement. Although 15 patients have died, only 4 deaths were because of metastatic choroidal melanoma. Of the original 48 patients, 33 (69%) have maintained within 2 lines or have better than their preoperative visual acuity. Side effects attributable to heating have included decreased intraocular pressure without hypotony as well as chorioretinal scar formation within and around the targeted zone.

Conclusions: The results of this series suggest that adjuvant microwave thermotherapy can be used with reduced doses of ophthalmic plaque radiation therapy to control the growth of uveal melanomas. Although the incidence of neovascular glaucoma, enucleation, and tumor regrowth is comparable to that of other series evaluating radiation alone, the visual acuities of microwave plaque thermotherapy-treated eyes were found to be superior. *Ophthalmology* 1997; 104:1794–1803

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Dr. Finger has no proprietary interest in the devices used in this study.

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